

Experiment 8: Monoalphabetic Substitution Cipher Using Keyword

```
#include <stdio.h>

#include <ctype.h>
#include <string.h>

int main()
{
    char text[100], key[] = "CIPHER";
    char cipher[26];
    int i, j, k = 0, used[26] = {0};

    for (i = 0; key[i] != '\0'; i++)
    {
        char ch = toupper(key[i]);

        if (!used[ch - 'A'])
        {
            cipher[k++] = ch;
            used[ch - 'A'] = 1;
        }
    }

    for (i = 0; i < 26; i++)
    {
        if (!used[i])
            cipher[k++] = 'A' + i;
    }

    printf("Enter plaintext: ");
```

```

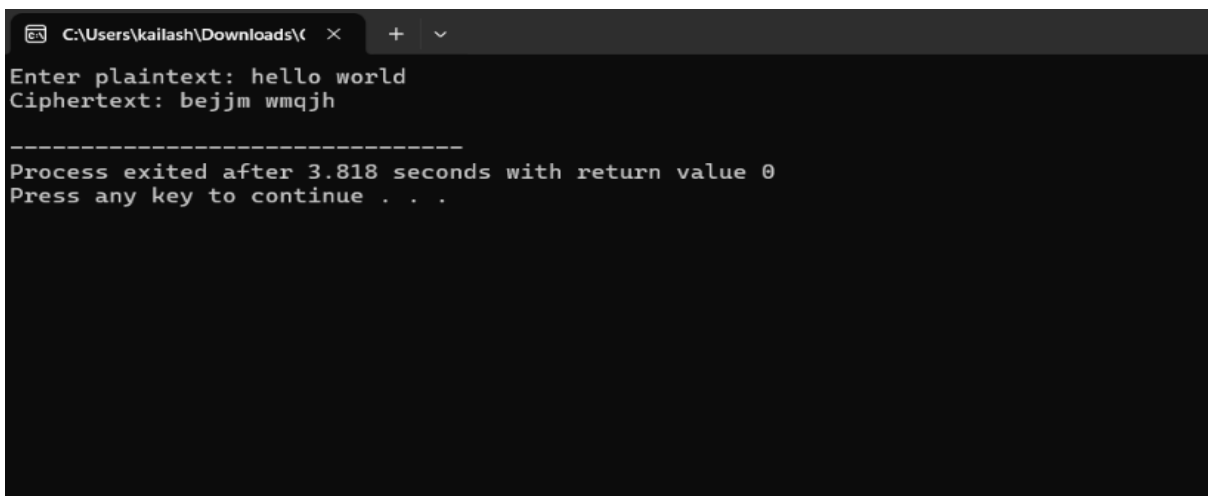
scanf(" %[^\\n]", text);

printf("Ciphertext: ");

for (i = 0; text[i] != '\\0'; i++)
{
    if (isalpha(text[i]))
    {
        j = toupper(text[i]) - 'A';
        printf("%c", islower(text[i]) ?
            tolower(cipher[j]) : cipher[j]);
    }
    else
    {
        printf("%c", text[i]);
    }
}
printf("\\n");
return 0;
}

```

Output:



```

C:\Users\kailash\Downloads\...
Enter plaintext: hello world
Ciphertext: bejjm wmqjh

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Process exited after 3.818 seconds with return value 0
Press any key to continue . . .

```